

APNASTAY

*Synopsis Report submitted in partial fulfillment
of the requirement for the degree of
B. E.(Information Technology)*

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CERTIFICATE OF APPROVAL

**For
Project Synopsis**

This is to Certify that

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APNASTAY

in partial fulfillment of degree course in
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We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

ApnaStay is an intelligent, mobile-first platform designed to simplify the process of relocating to a new city by integrating accommodation, roommate matching, and meal services into a single unified system. The primary objective of the platform is to eliminate the challenges faced by urban migrants by providing a centralized solution for essential living needs, thereby enhancing convenience, safety, and efficiency.

The system is designed to be user-friendly and highly interactive, allowing users to explore verified Paying Guest (PG) accommodations, find compatible roommates based on lifestyle preferences, and subscribe to home-cooked meal (dabba) services. Key features of the platform include AI-based roommate matching, image verification of property listings using computer vision techniques, and demand forecasting for meal services.

From a technical perspective, the platform utilizes Flutter for cross-platform mobile application development, while FastAPI is used to build a scalable backend system. PostgreSQL and Firebase are employed for efficient data management and authentication. Advanced Machine Learning models such as YOLO for image verification, NLP-based sentiment analysis for roommate matching, and LSTM for demand forecasting are integrated to enhance system intelligence and reliability. Secure payment transactions are handled through RazorPay integration.

The system is built using a microservices architecture, ensuring scalability, flexibility, and efficient performance. Initial testing demonstrates that ApnaStay effectively reduces the complexity of relocation by providing all essential services within a single platform.

Future enhancements include improved personalization using advanced AI models, expansion to more cities, and the addition of real-time features to further enhance user experience.

1. INTRODUCTION

Relocating to a new city is one of the most challenging transitions a person can face. Newcomers typically arrive with little to no local network, making it difficult to find reliable housing, nutritious food, and social companionship. The existing digital landscape provides fragmented and inconsistent tools users are forced to juggle multiple platforms to piece together even the most basic living arrangements.

ApnaStay is a proposed all-in-one mobile application designed specifically to address these pain points. By bringing together verified Paying Guest (PG) accommodation listings, intelligent roommate matching, and reliable home-cooked meal (dabba) services under a single unified platform, ApnaStay aims to make the relocation experience safe, efficient, and stress-free.

The platform leverages advanced Machine Learning (ML) algorithms, real-time data processing, and a modern microservices architecture to deliver a seamless, personalized user experience. ApnaStay targets students, working professionals, and anyone transitioning into an unfamiliar urban environment.

2. AIM AND OBJECTIVES

2.1 Aim

To develop a centralized, AI-powered mobile platform that simplifies urban relocation by providing integrated solutions for accommodation, roommate discovery, and meal services replacing the need for multiple disconnected applications.

2.2 Objectives

- **Centralized Convenience:** Build a unified platform serving as a one-stop solution for finding PG stays, compatible roommates, and daily meal services.
- **Algorithmic Roommate Matching:** Implement Collaborative Filtering combined with NLP sentiment analysis on user bios to generate accurate roommate compatibility scores.
- **Enhanced Verification & Safety:** Utilize a YOLO (You Only Look Once) computer vision model to verify property listing photos, and deploy anomaly detection algorithms to flag unhygienic food services.
- **Demand & Route Optimization:** Use LSTM Time Series Forecasting to predict service demand and Dijkstra/A* graph algorithms on Maps API data for optimal delivery route planning.
- **Secure & Scalable Architecture:** Employ a microservices backend using FastAPI, Firebase, JWT-based authentication, RBAC, and RazorPay integration to ensure a secure and production-ready system.

3. LITERATURE SURVEY

A review of existing solutions in the relocation and housing market reveals a highly fragmented landscape. Competing platforms address individual needs but fail to deliver a holistic solution:

- StayZ: Offers PG listings and roommate matching functionalities but entirely lacks meal integrations and broader daily life services. Users must rely on separate platforms for their food needs.
- Zolo: Provides PG listings and meal services but falls short by not offering roommate matching or personalized daily life services, leaving a critical social gap for newcomers.

The survey highlights a clear market gap: no existing platform combines all three pillars accommodation, social matching, and food into a single cohesive ecosystem. ApnaStay is specifically designed to bridge this gap.

Additionally, the following technical literature informed the design of ApnaStay's core ML features:

- Research papers on Collaborative Filtering and Content-Based Filtering for personalized recommendation systems.[1]
- Research papers on YOLO (You Only Look Once) real-time object detection for image verification pipelines.[3][4]
- Research on LSTM (Long Short-Term Memory) networks for time series forecasting and demand prediction.[5][6]
- Flutter official documentation for cross-platform mobile application development.[10]
- FastAPI official documentation for building high-performance Python-based API gateways.[11]

4. PROBLEM STATEMENT

When an individual relocates to an unfamiliar city whether for education, employment, or personal reasons they face an immediate, multi-layered crisis: they have no local contacts, no established living arrangements, and no access to reliable daily necessities. The current digital ecosystem forces them to navigate several separate, unverified, and inconsistent platforms to address each individual needs, resulting in significant time loss, financial risk, and emotional stress.

Specifically, the key problems identified are:

- **Fragmented Service Discovery:** There is no single platform where a newcomer can simultaneously search for verified PG accommodations, find a compatible roommate, and subscribe to a local meal service.
- **Lack of Verification & Trust:** Existing PG listing platforms lack robust photo and property verification, exposing users to fraudulent or misleading listings. Similarly, home-cooked meal providers have no quality-check mechanism.
- **Poor Roommate Compatibility:** Current platforms use simple filter-based matching rather than intelligent, preference-driven algorithms, leading to frequent incompatibility between roommates and high turnover.
- **Inefficient Meal & Service Logistics:** Home meal (dabba) services operate without demand forecasting or optimized delivery routing, resulting in inconsistent service quality and unnecessary delays.

Therefore, there exists a critical need for a unified, intelligent, and trustworthy platform that addresses all these challenges in one place. ApnaStay is proposed as the solution an end-to-end relocation companion that leverages machine learning, computer vision, and optimized logistics to make settling into a new city a safe, fast, and comfortable experience.

5. SCOPE

The scope of ApnaStay encompasses the following functional and technical boundaries:

5.1 In Scope

- **Mobile Application:** Cross-platform Flutter app supporting Android and iOS devices targeting urban migrants across major Indian cities.
- **PG Accommodation Module:** Listing creation, YOLO-based photo verification, geolocation search, filtering, and real-time availability tracking.
- **Roommate Matching Module:** User profile creation with lifestyle tags, bio-based NLP sentiment analysis, and Collaborative Filtering-based compatibility scoring.
- **Meal Service Module:** Subscription-based dabba service with LSTM demand forecasting, anomaly detection for hygiene flagging, and A*/Dijkstra route optimization.
- **Payment & Booking:** RazorPay integration for secure online payments, booking management, and digital receipts.
- **Admin Dashboard:** Web-based administration panel for platform moderation, user management, analytics, and report generation.

5.2 Out of Scope

- Physical property management or ownership transfer services.
- Real-time live video tours of PG properties.
- Legal documentation or rental agreement services.
- International relocation or cross-border services.
- Grocery or restaurant delivery services beyond home-cooked dabba meals.

6. PROPOSED SYSTEM

ApnaStay proposes a comprehensive mobile-first platform built on a robust microservices architecture. The system is designed with four primary modules integrated seamlessly through a unified API gateway:

6.1 User Authentication & Profile Management

A secure onboarding module supporting OTP-based phone verification, JWT token issuance, and Role-Based Access Control (RBAC) for Users, PG Owners, Meal Providers, and Admins. Firebase Authentication handles third-party OAuth logins (Google), and all sessions are managed via refresh token rotation.

6.2 PG Accommodation Module

PG owners can create detailed listings with geo-tagged location data, room types, amenities, pricing, and high-resolution photographs. Each submitted photo is processed by a YOLOv8 computer vision model to verify authenticity and flag stock or irrelevant images.[3][4] Users can search listings through an intelligent geospatial search engine with advanced filters.

6.3 Roommate Matching Module

Users complete a structured preference profile with lifestyle tags (sleep schedule, cleanliness, diet, habits) and a free-text bio. The matching engine applies Collaborative Filtering to identify users with statistically similar preference vectors, and NLP-based sentiment analysis on bio text to compute a holistic compatibility score.[1][2] Matched users are ranked and surfaced as recommendations.

6.4 Meal (Dabba) Service Module

Home-kitchen providers can register and list their daily menus with pricing, delivery area, and availability schedules. An LSTM time series model continuously forecasts demand by locality to help providers plan capacity.[5][6] Order assignments are optimized through Dijkstra/A* graph algorithms applied on real-time Maps API data to compute minimum-cost delivery routes.[8][9][17]

7. METHODOLOGY

7.1 Overall Development Approach

ApnaStay follows an Agile-Scrum development methodology with iterative 2-week sprints. Each sprint delivers a tested, potentially shippable increment. The project is divided into four primary phases: Requirements & Design, Backend Development, Frontend Development, and Integration & Testing.

7.2 Machine Learning Pipeline

- Collaborative Filtering (Roommate Matching): User preference vectors are constructed from lifestyle tags. Cosine similarity between vectors produces pairwise compatibility scores. NLP (BERT-based sentiment analysis) is applied on bio text to produce a sentiment score, and the final compatibility score is a weighted combination of both.[2]
- YOLO Photo Verification: YOLOv8 is trained on a labelled dataset of legitimate PG property photographs versus stock/inappropriate images.[3][4] Each submitted listing photo is run through the inference pipeline; flagged images trigger a review workflow.
- LSTM Demand Forecasting: Historical order data (by locality, day-of-week, time-of-day) is used to train an LSTM sequence model.[5][6] The model predicts next-day demand volume for each service area, enabling providers to plan meal preparation capacity accordingly.
- Anomaly Detection (Meal Hygiene): Statistical anomaly detection algorithms (Isolation Forest) are applied to provider review data and hygiene score time series to automatically flag deteriorating service quality for admin review.[7]
- Route Optimization: Dijkstra's algorithm with A* heuristic is applied on a weighted graph derived from the Google Maps API to compute optimal delivery routes for dabba providers, minimizing total delivery distance and time.[8][9][17]

7.3 Backend Architecture

The backend employs a microservices architecture with the following independently deployed services: Auth Service, Listing Service, Matching Service, Meal Service, Payment Service, and Notification Service. All services communicate through a FastAPI Gateway using REST APIs. Redis is used for caching session data and ML inference results.

7.4 Frontend Architecture

The Flutter-based mobile application follows the BLoC (Business Logic Component) state management pattern. The UI is built with Material Design 3 components and supports both Android and iOS. The app communicates exclusively with the API Gateway; direct database access is not permitted from the client.

8. ANALYSIS

8.1 Process Model Used for the Project

The development of the ApnaStay system follows the Agile Scrum Model, which is an iterative and incremental approach to software development. This model is chosen due to its flexibility, adaptability, and ability to incorporate continuous feedback throughout the development lifecycle.

The project is divided into multiple short development cycles called sprints, each typically lasting two weeks. During each sprint, specific modules such as user authentication, PG listing, roommate matching, and meal services are developed, tested, and reviewed.

Agile methodology allows:

- Continuous interaction with stakeholders
- Early detection and resolution of issues
- Incremental delivery of functional components
- Better management of changing requirements

This approach ensures that the ApnaStay platform evolves efficiently while maintaining high quality and user satisfaction.

8.2 Feasibility Study

The feasibility of the ApnaStay system is analyzed from three major perspectives:

8.2.1 Technical Feasibility

The system is technically feasible as it utilizes well-established and widely supported technologies such as Flutter, FastAPI, PostgreSQL, Firebase, and Machine Learning frameworks.

The integration of advanced technologies like:

- YOLO for image verification
- LSTM for demand forecasting
- NLP for roommate matching

is achievable using existing libraries and tools. The development team has access to the required technical expertise, making the implementation practical and achievable.

8.2.2 Economic Feasibility

The project is economically feasible as it primarily relies on open-source technologies, reducing software licensing costs.

Costs involved include:

- Cloud hosting and database services
- API usage (Google Maps, RazorPay)
- Maintenance and updates

Since the platform has potential for scalability and monetization (through bookings and subscriptions), the overall cost is justified and sustainable.

8.2.3 Operational Feasibility

The system is user-friendly and designed to meet the needs of students and working professionals relocating to new cities.

The mobile application provides:

- Simple navigation
- Easy search and booking features
- Integrated services in one platform

Users do not require any special training, making the system operationally feasible. Additionally, admins can efficiently manage users, listings, and services through a centralized dashboard

8.3 Cost Analysis

The cost structure of the ApnaStay system is divided into the following components:

- Development Cost: Minimal, as open-source tools and frameworks are used
- Infrastructure Cost: Includes cloud hosting, database services, and API integrations
- Maintenance Cost: Regular updates, bug fixes, and monitoring
- Operational Cost: Server scaling, analytics processing, and ML model updates

Overall, the system is cost-effective due to its reliance on scalable and cloud-based resources.

8.4 Timeline Chart (For the Entire Project)

The development of the ApnaStay system is planned over a duration of 8 months, following an Agile-based approach with iterative development and continuous feedback.

The project is divided into the following phases:

Phase	Description	Duration
Requirement Analysis	Understanding user needs, defining system scope and problem statement	1 Month
System Design	Creating system architecture, DFDs, UML, and database design	1 Month
Backend Development	Development of APIs, database integration, authentication system	2 Months
Frontend Development	Development of Flutter mobile application and UI/UX design	2 Months
Machine Learning Integration	Implementation of YOLO, NLP, LSTM models and testing	1 Month
Integration & Testing	System integration, debugging, and performance testing	1 Month

9. DESIGN

9.1 Data Flow Diagrams

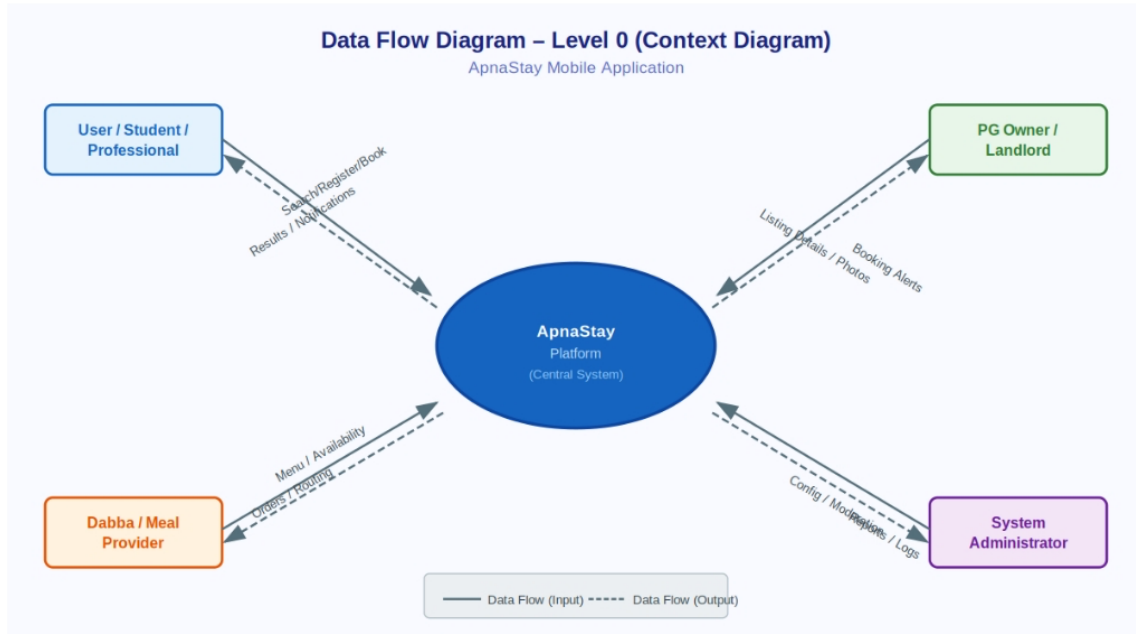


Fig.9.1: Data Flow Diagram – Level 0 (Context Diagram)

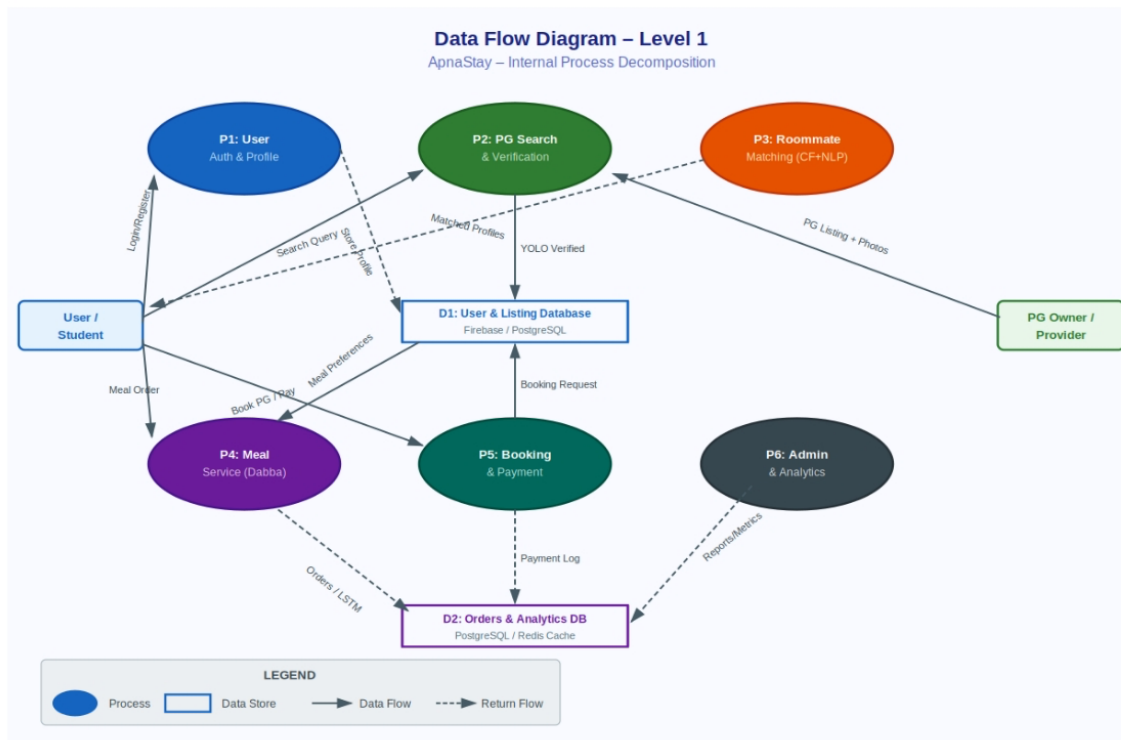


Fig.9.2: Data Flow Diagram – Level 1

9.2 UML Diagrams

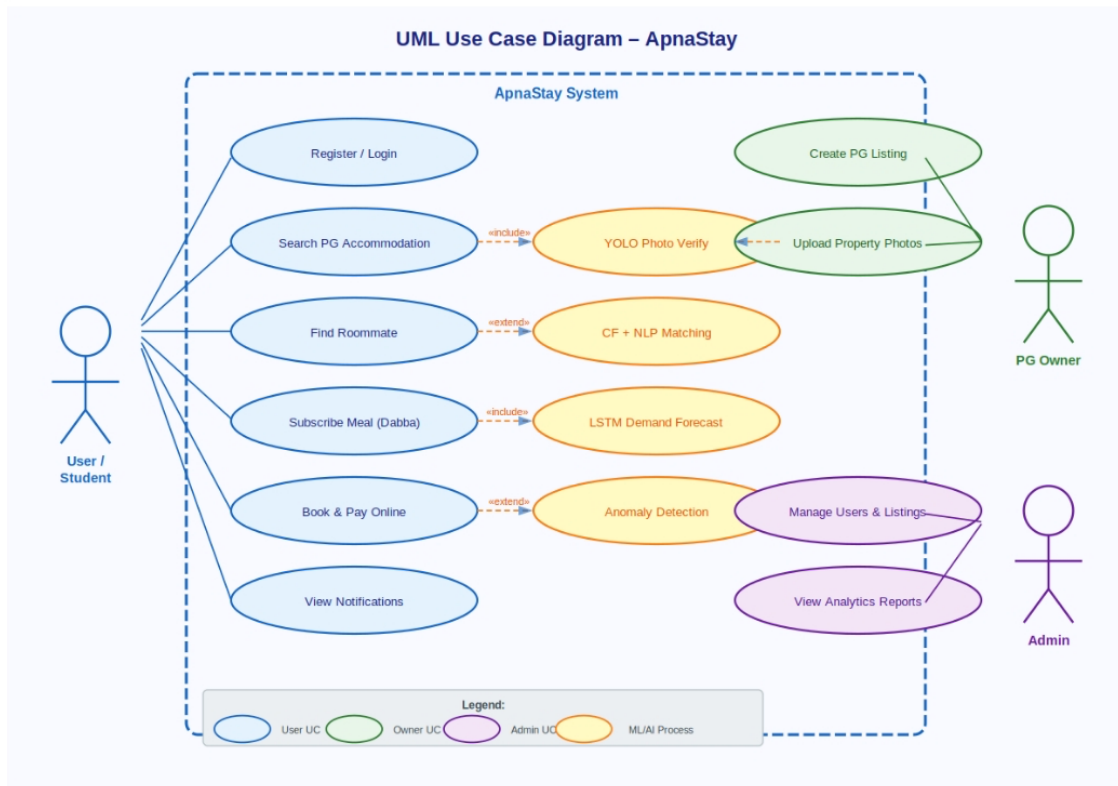


Figure 9.3: Use Case Diagram showing interactions between Users, PG Owners, Admins, and the ApnaStay system processes.

Fig.9.3: UML Use Case Diagram

9.3 EER Diagram

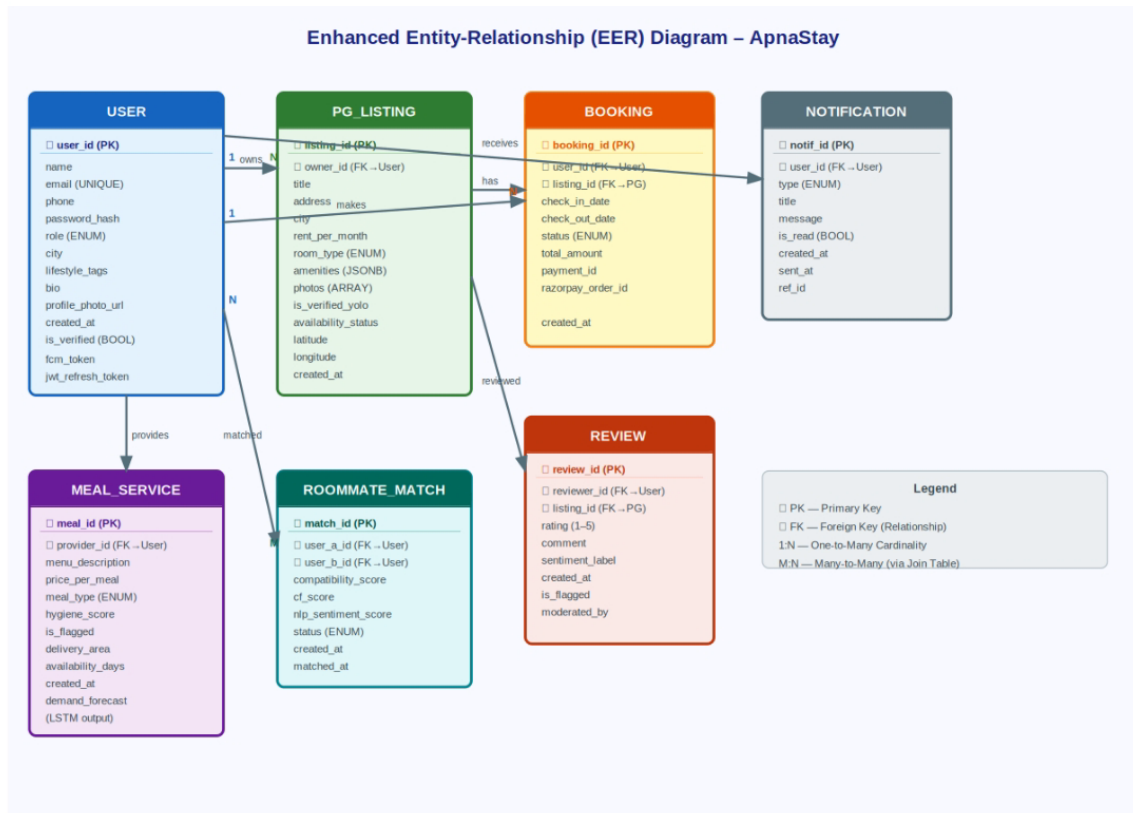


Figure 9.4: Enhanced Entity-Relationship Diagram showing all entities, attributes, primary/foreign keys, and relationships.

Fig.9.4: Enhanced Entity Relationship Diagram

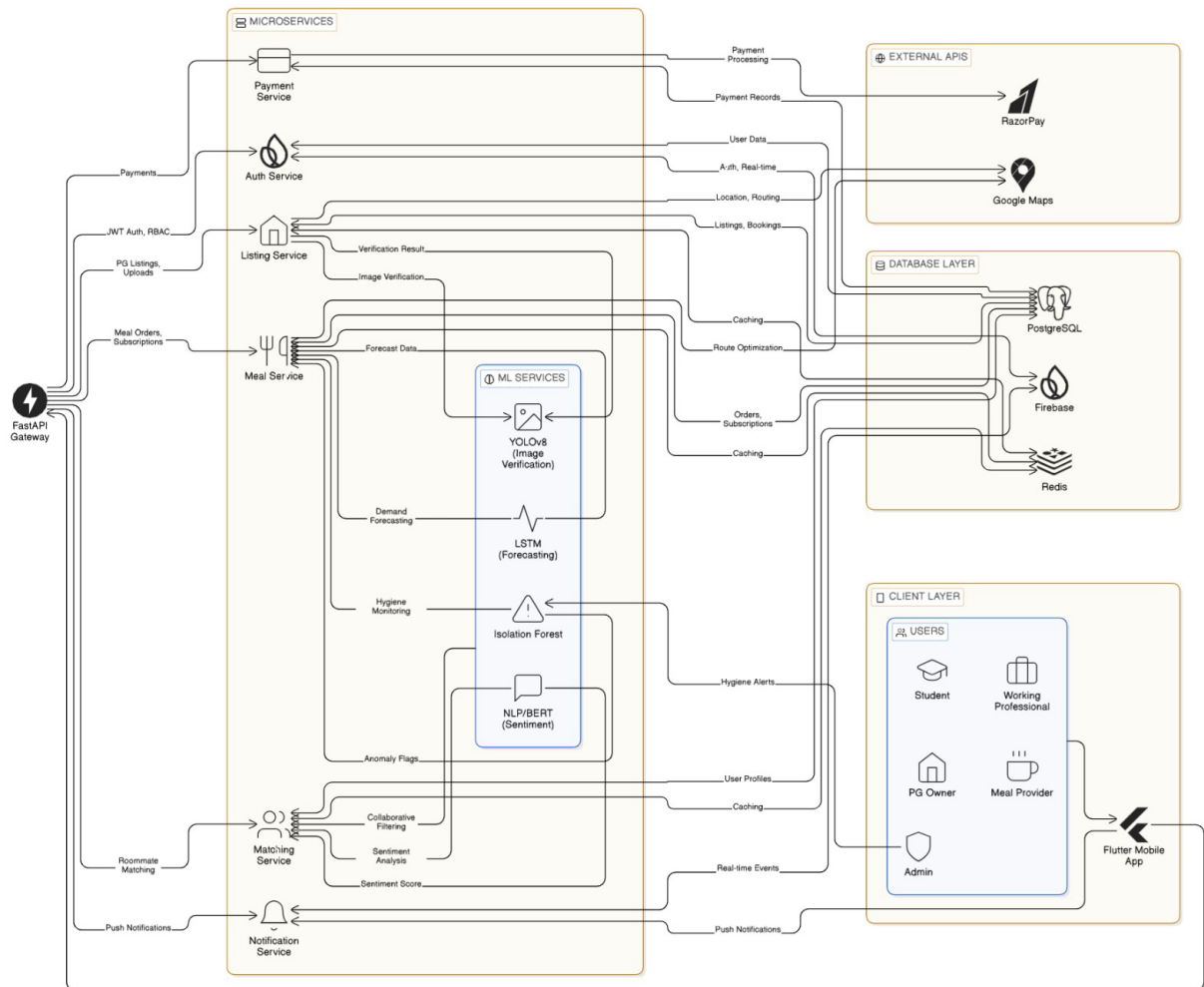


Fig.9.5: Architecture Diagram

10. HARDWARE AND SOFTWARE REQUIREMENTS

10.1 Hardware Requirements

The development and deployment of the ApnaStay system require standard computing hardware capable of supporting mobile application development, backend services, and machine learning model execution.

The minimum hardware requirements are as follows:

- Processor: Intel Core i5 or higher
- RAM: Minimum 8 GB (16 GB recommended for ML processing)
- Storage: Minimum 256 GB SSD
- Device: Android/iOS smartphone for testing the mobile application
- Internet Connectivity: Stable broadband connection for API integration and cloud services

These specifications are sufficient to handle development tasks such as coding, database management, API testing, and machine learning model training.

10.2 Software Requirements (Tech Stack)

The ApnaStay system is built using modern, scalable, and open-source technologies across different layers of the application:

Frontend

- Flutter: Used for developing a cross-platform mobile application supporting both Android and iOS devices[10]
- Dart: Programming language used in Flutter

Backend

- FastAPI: High-performance Python framework used for building RESTful APIs[11]
- Python: Core programming language for backend and ML integration

Database

- PostgreSQL: Used for structured data storage (users, listings, bookings) [12]
- Firebase: Used for authentication and real-time services [13]

Machine Learning & AI

- TensorFlow / PyTorch: Frameworks used for building and training ML models [14][15]
- YOLOv8: Used for image verification of PG listings [3][4]
- NLP (BERT-based models): Used for roommate matching analysis [2]
- LSTM: Used for demand forecasting [5][6]

APIs and Integrations

- RazorPay: Payment gateway integration for secure transactions [16]
- Google Maps API: Used for geolocation and route optimization [17]

Other Tools & Technologies

- Redis: Used for caching and improving system performance
- Git & GitHub: Version control and code management
- Postman: API testing
- Docker (optional): Containerization for deployment

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